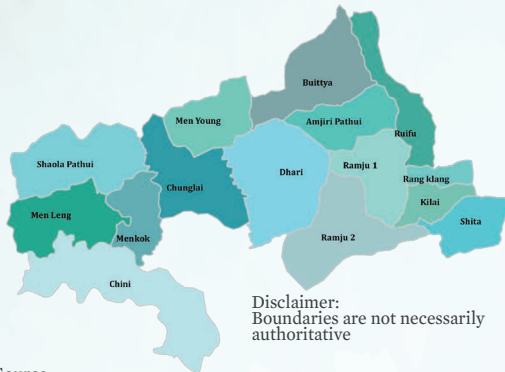


Tangkabati 2 Watershed

B-77

Figure 1:
Para (village) boundary of Tangkabati 2 watershed



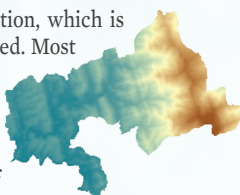
Source:
Center for Environmental and
Geographic Information Services (CEGIS)

The Tangkabati 2 sub-watershed (B-77), located in the Tangkabati Union, 309 no Mauza of the Bandarban District, encompasses 15 paras (figure 1) and covers approximately 1,567 hectares of land. It is home to almost 2,000 people, with communities consisting of the Mro ethnic group only. The overall literacy rate is approximately 58%, yet opportunities for higher education are very limited. Farming is the main occupation. The average monthly income is 11,735 taka (\$98), which is relatively low compared to the national average. About 88% of the houses are katcha and almost 94% of households use solar electricity systems. There is a noticeable scarcity of drinking water and a prevalent incidence of waterborne diseases.

Topography

Only 1% area lies below 40-meter (m) elevation, which is situated in the western region of the watershed. Most of the study area lies between 40–120m (36%) and 120–300m (28%) elevation. The hilly ridge along the east of the watershed has comparatively higher elevation but only a little area falls under this class. About 21% of the area lies between 300–500m elevation and 12% of the area lies between 500–650m elevation. Only 1% of the area lies above 650m elevation, which is situated in the eastern hilly region of the watershed.

Figure 2: Topography



Source:
Center for Environmental and
Geographic Information Services (CEGIS)

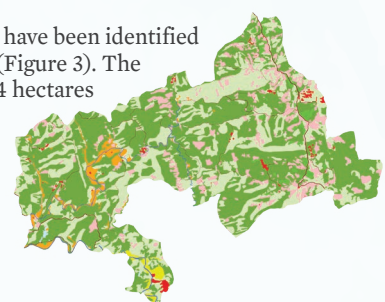
Disclaimer:
Boundaries are not
necessarily authoritative



Landuse

A total 12 types of land use have been identified within the B77 watershed (Figure 3). The main land use is forest (834 hectares (ha)), followed by herb/shrub (478 ha), shifting cultivation (149 ha), and cropland (39 ha).

Figure 3: Land use

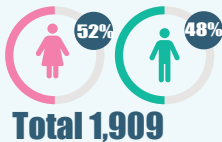


Disclaimer:
Boundaries are not necessarily
authoritative

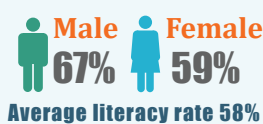
Source:
Center for Environmental and
Geographic Information Services (CEGIS)

Demography

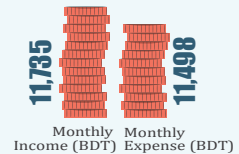
Population



Literacy



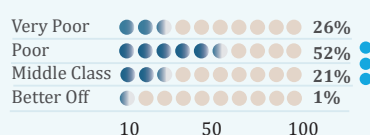
Economy



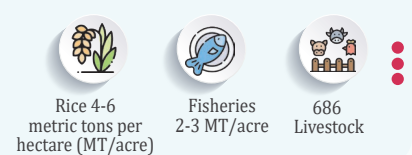
WASH



Poverty



Production



Climatic Hazards



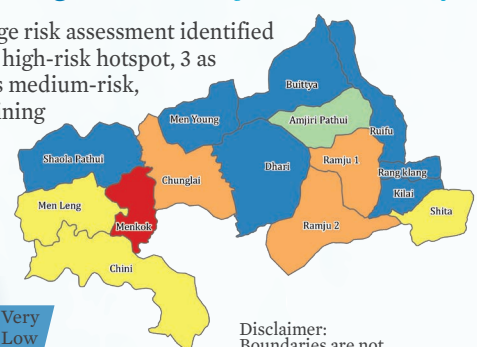
Climate Change-Induced Hazards

Nine climate hazards threaten Tangkabati 2 watershed agriculture, livelihood, ecosystem, and biodiversity, drought, erratic rainfall, and landslides intensify across most of the paras. Heat waves are increasing with high intensity in most of the paras. Flash floods, river floods, lightning, and tornadoes occur at low to moderate intensity.

Climate Change Risk

Climate change risk assessment identified 1 para as very high-risk hotspot, 3 as high-risk, 3 as medium-risk, and the remaining paras as low to very low-risk (figure 4).

Figure 4: Mutihazard risk map

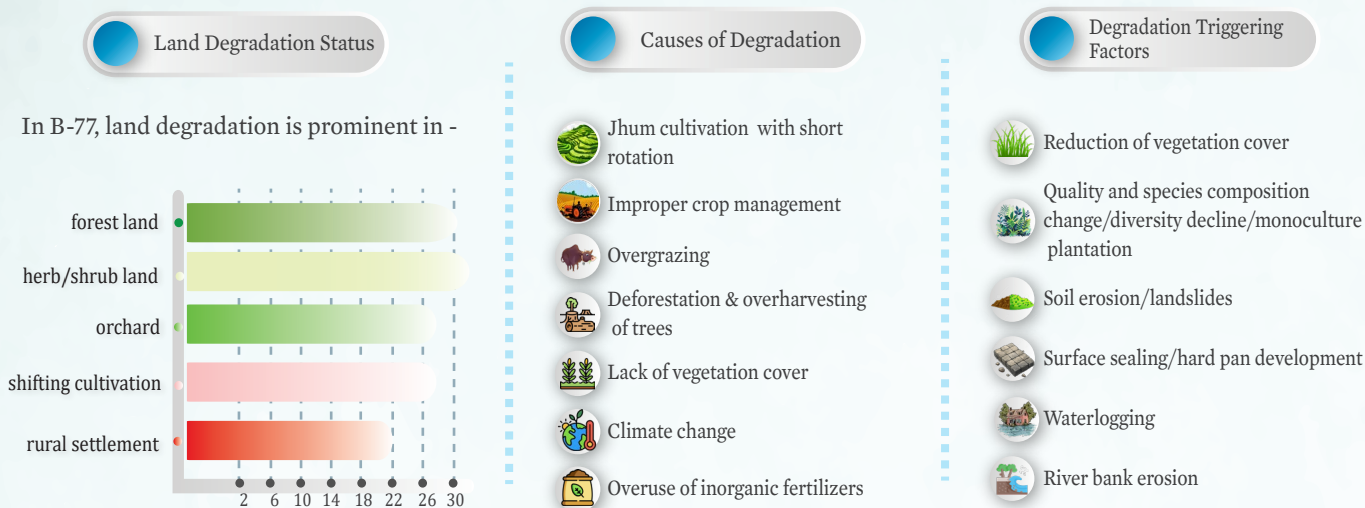


Disclaimer:
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Watershed Degradation

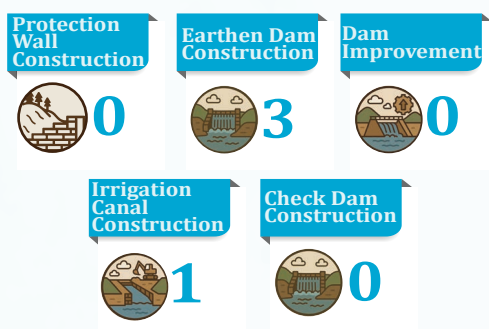


Watershed Conservation

Conservation measures focused on enhancing ecological resilience as well as improving livelihood quality of the local community were identified and categorized into 6 classes.



Watershed Management Infrastructures



Watershed conservation interventions in Tangkabati 2 include dam construction (3) and irrigation canal construction (1) being the most implemented, aimed at enhancing water storage and distribution. In addition to these, other watershed conservation measures have been identified to further boost water availability, agricultural productivity, and overall ecosystem health. Measures such as check dam construction, dam improvement, irrigation canal construction, culvert construction, and protection wall construction will help control runoff, reduce soil erosion, and protect critical infrastructure. Together, these interventions will significantly benefit the local community by ensuring year-round water access, improving crop yields, and reducing vulnerability to climate-induced hazards. These integrated efforts promote sustainable livelihoods and strengthen resilience in the Tangkabati 2 region.